VCE Unit 4
SAC 1: ORGANIC COMPOUNDS AND SPECTROSCOPY ANALYSIS

NAME: Marking Scheme **Total Marks 50**

TEACHER (circle one): **MC** **RH** **JJ** **SP** **NP**

Questions 1 and 2 relate to this list of organic compounds

i. $CH_3CH_2COCH_2CHO$
ii. $CH_3CH_2CH_2CH_2COOH$
iii. $CH_3COCH(CH_3)CHO$
iv. $CH_2CHCH_2CH_2COOH$

1. Which of the organic compounds listed above are isomers of $C_5H_8O_2$?

A. i and ii only
B. i, ii and iv only
C. i, iii and iv only
D. ii, iii and iv only

2. Which of the organic compounds listed above contains a chiral centre?

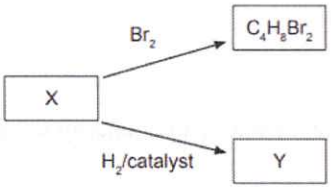
A. i only
B. i, iii only
C. iii only
D. ii, iii only

3. Which one of the following lists the substances in order of increasing (from lowest to highest) boiling point?

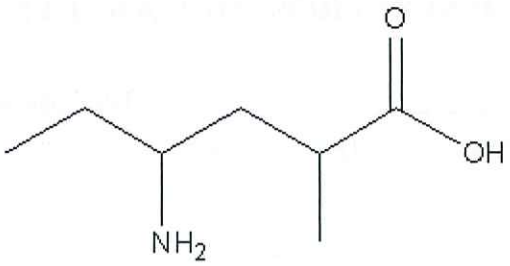
A. Ethane, ethanol, ethanal, ethanoic acid
B. Ethane, ethanal, ethanol, ethanoic acid
C. Ethanol, ethane, ethanoic acid, ethanal
D. Ethanoic acid, ethanal, ethanol, ethane

4. In the diagram shown below, what are the molecular formulae of substance X and Y likely to be?

	X	Y
A.	C_4H_8	C_4H_6
B.	C_4H_8	C_4H_{10}
C.	C_4H_{10}	C_4H_8
D.	C_4H_6	C_4H_8



5. What is the IUPAC name for the following compound?



A. 3-amino-5-methylhexanal
B. 4-amino-2-methylhexanoic acid
C. 3-amino-5-methylhexanoic acid
D. 4-amino-2-methylhexanal

6. The table below lists some of the properties of the straight chained carbon compounds *W*, *X*, *Y*, *Z*.

Compound	Reactivity in bromine water	Solubility in water
W	Rapidly decolourises	Insoluble
X	Unreactive	Insoluble
Y	Unreactive	Soluble
Z	Unreactive	Partly soluble

Which row of the following table best identifies the compounds *W*, *X*, *Y* and *Z*?

	W	*X*	*Y*	*Z*
A.	C_3H_6	C_3H_8	CH_3OH	C_4H_9OH
B.	C_3H_8	C_3H_6	CH_3OH	C_4H_9OH
C.	C_3H_6	C_3H_8	C_4H_9OH	CH_3OH
D.	C_3H_8	C_3H_6	C_4H_9OH	CH_3OH

7. Which compound is a member of the aldehyde (alkanal) homologous series?

A. CH_3CH_2OH
B. CH_3CH_2COOH
C. CH_3COCH_3
D. CH_3CH_2CHO

8. Which spectroscopic technique uses energy to cause vibrations and stretching of bonds?

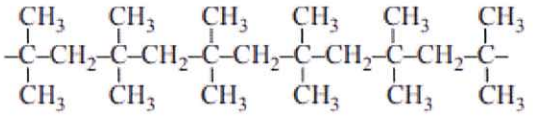
A. Proton NMR
B. Mass Spectrometry
C. Flame test
D. Infrared Spectroscopy

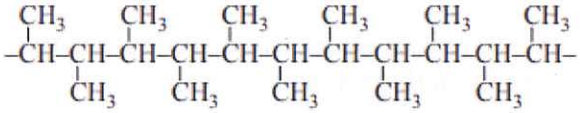
9. A sample of a hydrocarbon for analysis is placed in a strong magnetic field and irradiated with electromagnetic radiation of radio wave frequency.

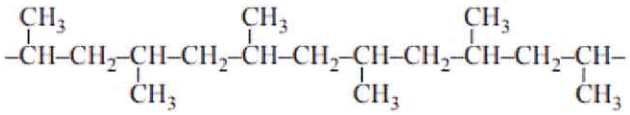
This is most likely to result in

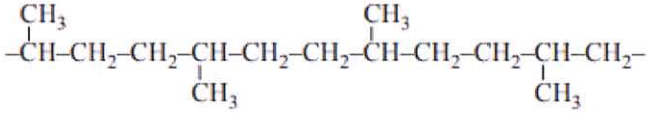
A. ionisation and fragmentation of molecules.
B. an increase in bond vibrations of molecules.
C. the promotion of electrons to higher energy levels.
D. a change in the energy of the nuclei of the hydrogen atoms.

10. Consider the addition polymerisation of $CH_3CH{=}CHCH_3$. The structure of the resulting polymer would be

A.


B.


C.


D.


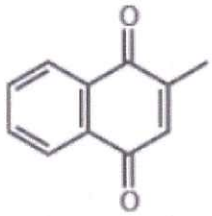
11. How many hydrogen atoms are there in a molecule of octan-3-ol?

A. 8
B. 18
C. 19
D. 20

Menadione is a molecule that forms the basic structure of Vitamin K molecules. Vitamin K helps the blood clot and is needed for good bone health. Questions 12 and 13 relate to the structure of Menadione shown here.

12. The molar mass, in g mol^{-1}, of menadione is closest to:

A. 157
B. 160
C. 171
D. 172



MENADIONE

13. What is the name of the functional group featured twice in this structure?

A. Carboxyl/ Carboxy
B. Carbonyl (ketone)
C. Carbonyl (aldehyde)
D. Hydroxyl/ Hydroxy

14. For which one of the following molecular formulae is there only one possible structure?

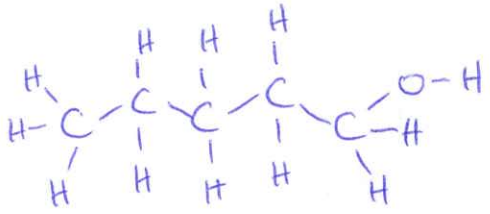
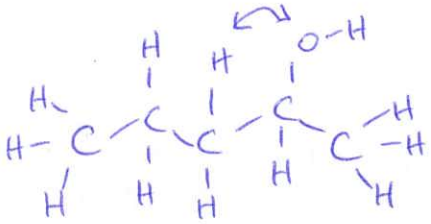
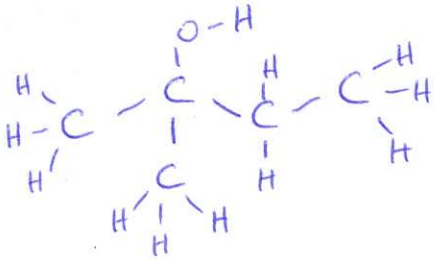
A. $C_4H_9NH_2$
B. $C_2H_2Br_2$
C. $C_2H_4Br_2$
D. C_2HBr_3

15. In which of the following molecules do geometric isomers **NOT** exist?

A. 3-ethylhex-3-ene
B. but-2-ene
C. pent-2-ene
D. 5-methylhept-3-ene

16. Complete the table below by drawing the structure for a primary alcohol, a secondary alcohol and a tertiary alcohol, each with the molecular formula $C_5H_{12}O$. **Show all H atoms** in your structures.

Give the IUPAC names for the primary and secondary alcohols you have drawn.

	Structure	Name
Primary alcohol		pentan-1-ol
Secondary alcohol		pentan-2-ol or pentan-3-ol
Tertiary alcohol		**Name not required.**

5 marks

17. Complete the following table:

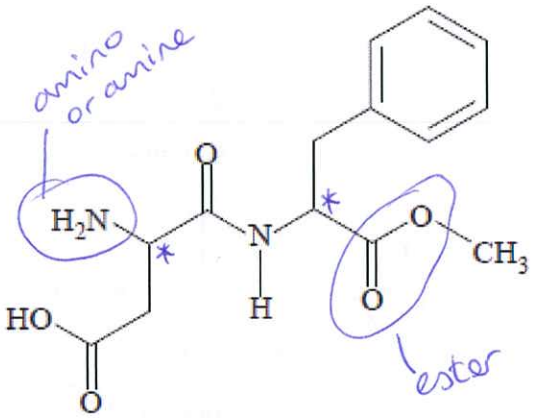
Structural Name (IUPAC)	Semi-Structural formula
2–bromo–3–methylbuta-1-ene	$H_2CCBrCH(CH_3)_2$
3-hydroxypropanoic acid	$CH_2(OH)CH_2COOH$
5-aminohexan-2-ol	$CH_3CH(NH_2)CH_2CH_2CH(OH)CH_3$
propylpentanoate	$H_3C(CH_2)_2OCO(CH_2)_3CH_3$

4 marks

18. Aspartame is an artificial sweetener that was accidently discovered in 1965 as a pharmaceutical company sought a new way to treat ulcers. Aspartame is the methyl ester of a dipeptide formed by the condensation reaction between the amino acids Asp and Phe.

On the aspartame molecule shown below:

a) Circle and label the **amine** functional group
b) Label **all chiral centres** with an asterix (*)
c) Circle and label the **ester** functional group
d) State whether the amide functional group is primary, secondary or tertiary. secondary



aspartame

5 marks

19. Alkenes can be used to manufacture a range of products. The reaction pathway shown below represents one example of the use of an alkene.
In this reaction pathway, Compound P is used to produce Compound R and Compound S. Compound S can then be used to produce Compound T.

Complete the following in the appropriate boxes in the reaction pathway diagram provided.

a. Give the IUPAC systematic name for Compound P and Compound R.
b. Write the formulas of the reagent and the catalyst required to produce Compound R and Compound S from Compound P.
c. Write the **semi-structural** formula of Compound S.
d. Draw the **structural** formula of Compound T, showing all bonds.

Compound P

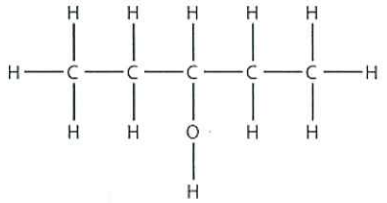
$CH_3CH_2CHCHCH_3$

Name pent-2-ene

Reagent H_2O (g)

Catalyst H_3PO_4 or H^+

Compound R



Name pentan-3-ol

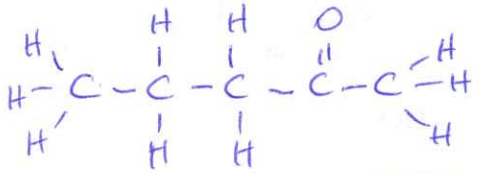
Compound S

$H_3C(CH_2)_2CH(OH)CH_3$

Semi-structural formula

$Cr_2O_7^{2-}(aq)/H^+(aq)$

Compound T



Structural formula (show all bonds)

2+1+1+1=5 marks

20. The Hill's Synthetics manufacturing company can produce ethanoic acid, $C_2H_4O_2$, by two different processes.

Process 1 $C_2H_6O + O_2 \rightarrow C_2H_4O_2 + H_2O$

Process 2 $CH_4O + CO \rightarrow C_2H_4O_2$

Use the table below for relevant molar masses.

Compound	**Formula**	**M (g mol^{-1})**
Ethanol	C_2H_6O	46
Oxygen	O_2	32
Ethanoic acid	$C_2H_4O_2$	60
Water	H_2O	18
Methanol	CH_4O	32
Carbon monoxide	CO	28

a. Hill's Synthetics know that the atom economy of a process is important.

Calculate the percentage atom economy for making ethanoic acid by process 1.

$\frac{60}{46+32} \times 100$

$= 77\%$

b. Hill's Synthetics also know that the percentage yield of a process is important.
The factory uses 5.2 tonnes of methanol in process 2.
A chemist predicts they should make 9.8 tonnes of ethanoic acid. They actually make 8.5 tonnes of ethanoic acid.
Calculate the percentage yield of the ethanoic acid.

$n(CH_4O) = \frac{5.2 \times 10^6}{32}$

$= 1.6 \times 10^5$ mol

$n(CH_4O) = n(C_2H_4O_2)$

$m(C_2H_4O_2) = 1.6 \times 10^4 \times 60 = 9.8 \times 10^6$ g

$\frac{8.5}{9.8} \times 100 = 87\%$

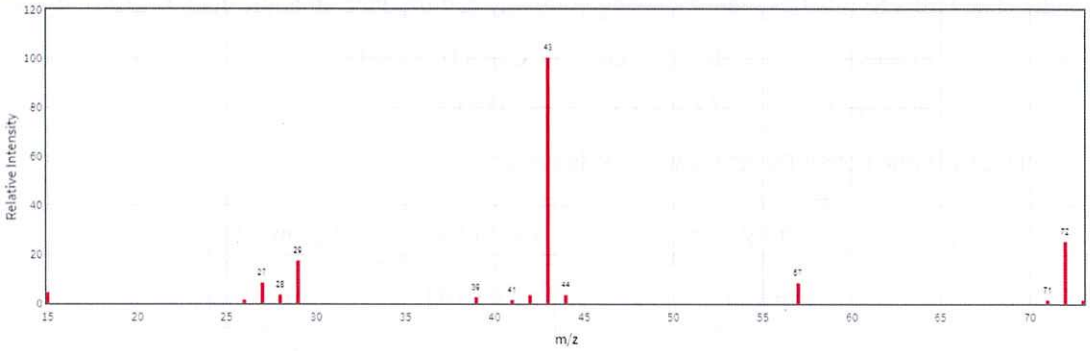
c. Other than cost, what is one advantage of a very high atom economy?

No waste to dispose of.

1+1+1=3 marks

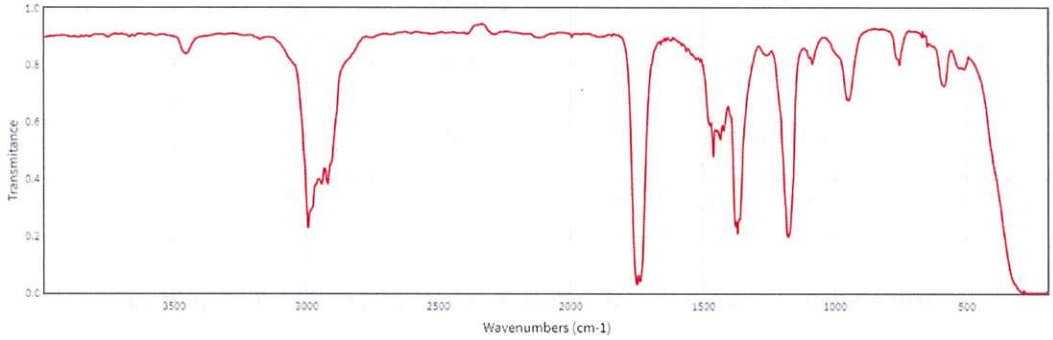
21. The four spectra shown below are for a compound with the empirical formula C_4H_8O (M=72).

Mass Spectrum



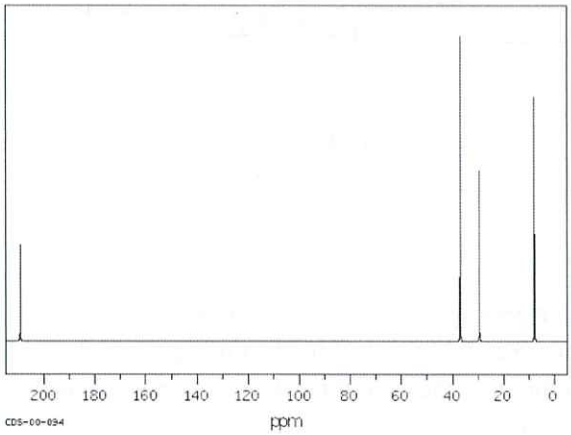
Source: https://webbook.nist.gov/cgi/cbook.cgi?ID=C78933&Units=SI&Mask=200#Mass-Spec Accessed 25/6/18

Infrared Spectrum



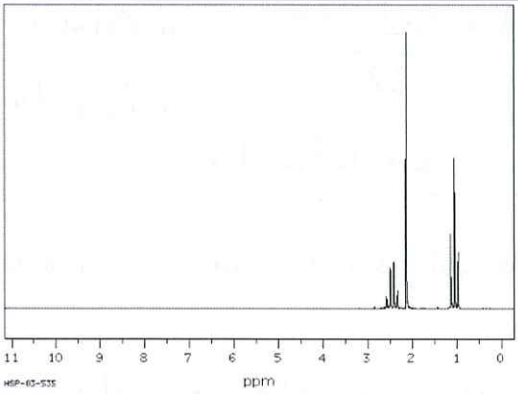
Source: https://webbook.nist.gov/cgi/cbook.cgi?ID=C78933&Units=SI&Type=IR-SPEC&Index=1#IR-SPEC Accessed 25/6/18

^{13}C NMR Spectrum



Source: https://webbook.nist.gov/cgi/cbook.cgi?ID=C78933&Units=SI&Type=IR-SPEC&Index=1#IR-SPEC Accessed 25/6/18

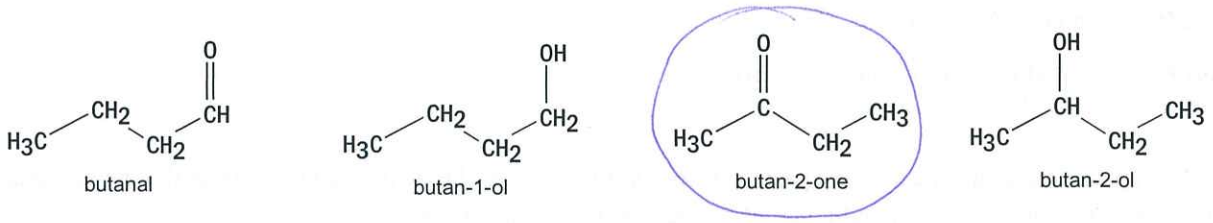
^{1}H NMR Spectrum



Source: https://www.chemicalbook.com/SpectrumEN_78-93-3_1HNMR.htm Accessed 25/6/18

Use the spectra to deduce the structure of the compound, and explain how each spectrum supports your proposed structure by using **two main pieces of evidence** from each spectra.

a. Circle the correct structure of the unknown compound.



Any 2 of:

b. Mass spectrum information.

- molecular ion m/z 72
- m/z 43 (base peak) could be CH_3CO^+ or CH_2CHO^+

c. Infrared spectrum information.

- stretch @ ≃ 1700 C=O
- No O-H stretch
- Stretch @ ≃ 3000 C-H

d. ^{13}C NMR spectrum information.

- 4 carbon environments
- peak @ 210 $R_2C=O$

e. ^{1}H NMR spectrum information.

- 3 hydrogen environments
- peak @ 2.5 is a quartet ∴ next to CH_3
- aldehyde ruled out by absence of large shift.

1+2+2+2+2=9 marks

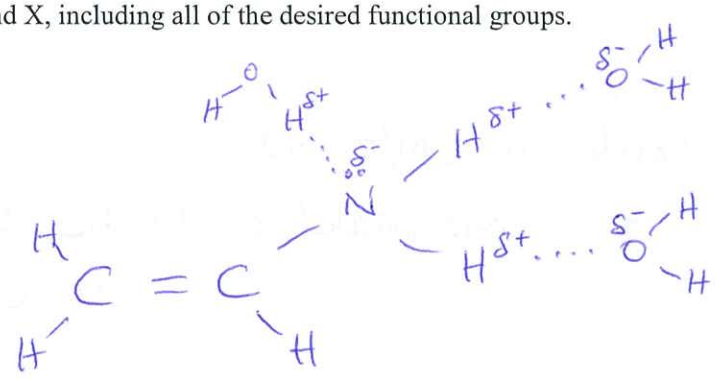
22. Kerry performed a synthesis to form a new compound. She called the compound "Compound X". In order to ensure that the synthesis had produced the desired functional groups, Kerry performed the following tests. She recorded her observations below.

Test A: Four drops of bromine were added to compound X and the brown colour disappeared.

Test B: Compound X is soluble in water.

Test C: The solution of X in water had a pH of 8.5

a) Mass spectroscopy determined the parent molecular ion to be 43. Draw the structural formula, including any lone pairs, of Compound X, including all of the desired functional groups.



* C=C
* NH_2
* correctly drawn.

b) On your drawing of Compound X, show where all the hydrogen bonds with water would form by drawing water molecules.

* 3 water molecules shown correctly.

4 marks

End of Task – 50 marks available